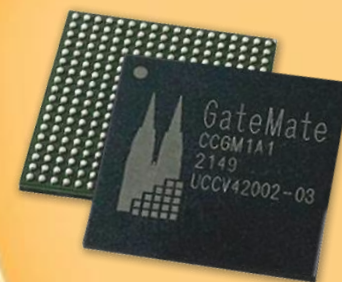


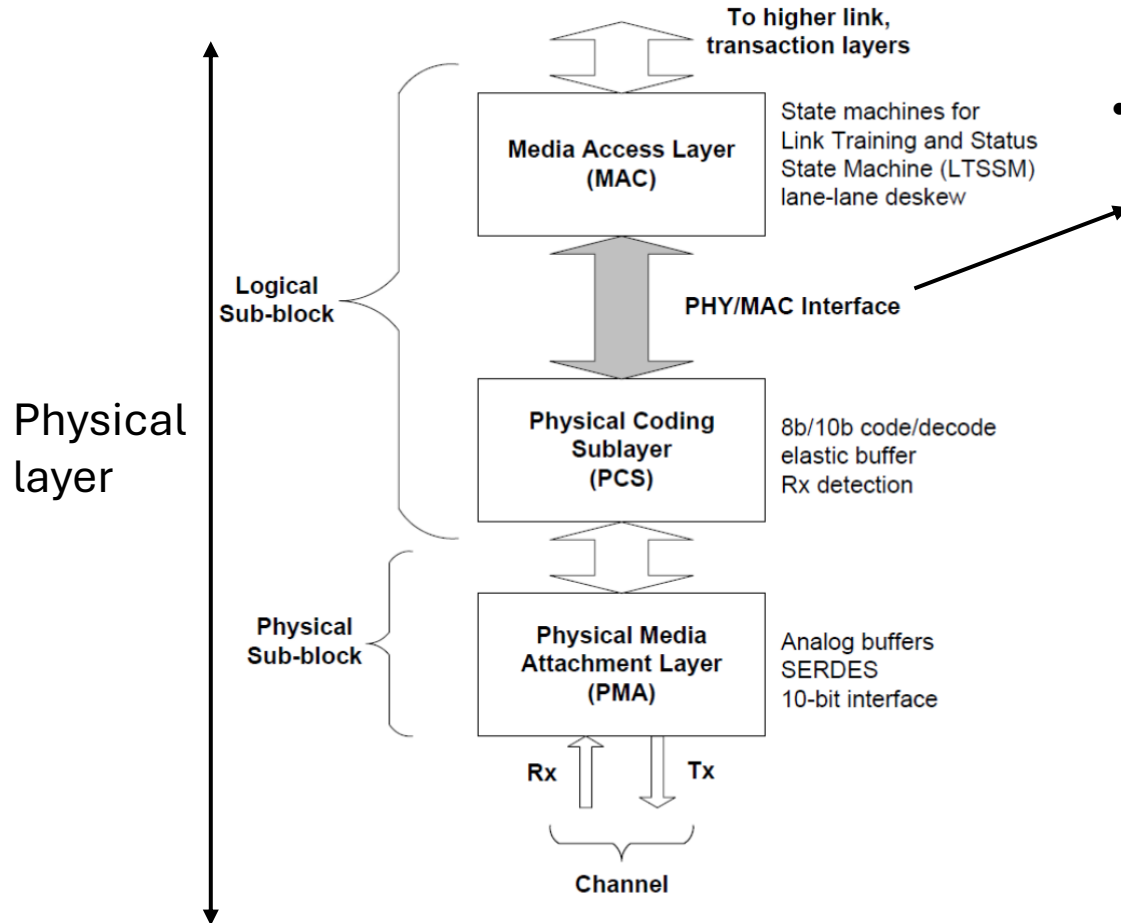
GateMate™ FPGA

PHY Interface for the PCIe Architecture

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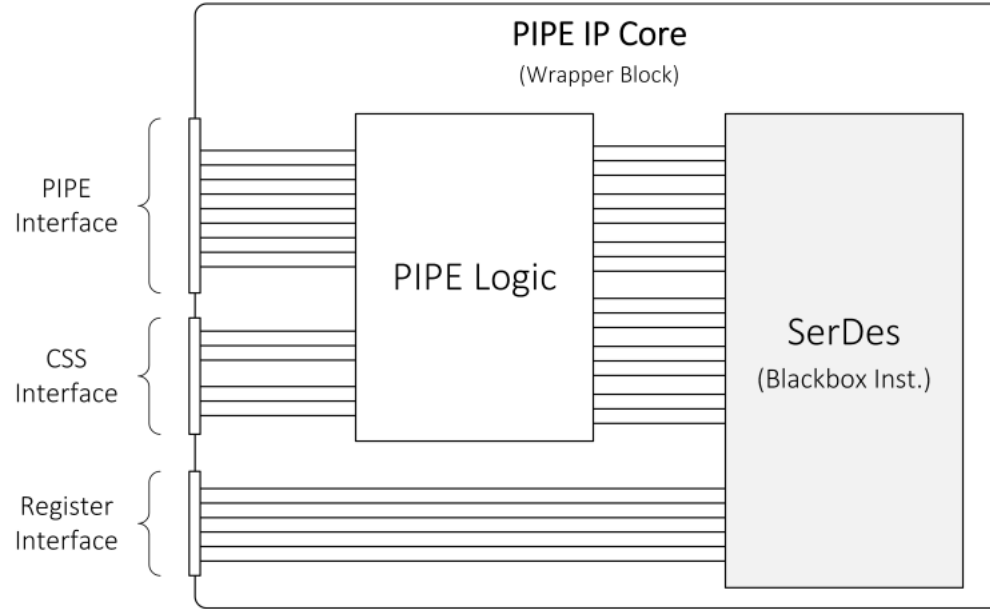


PIPE Overview



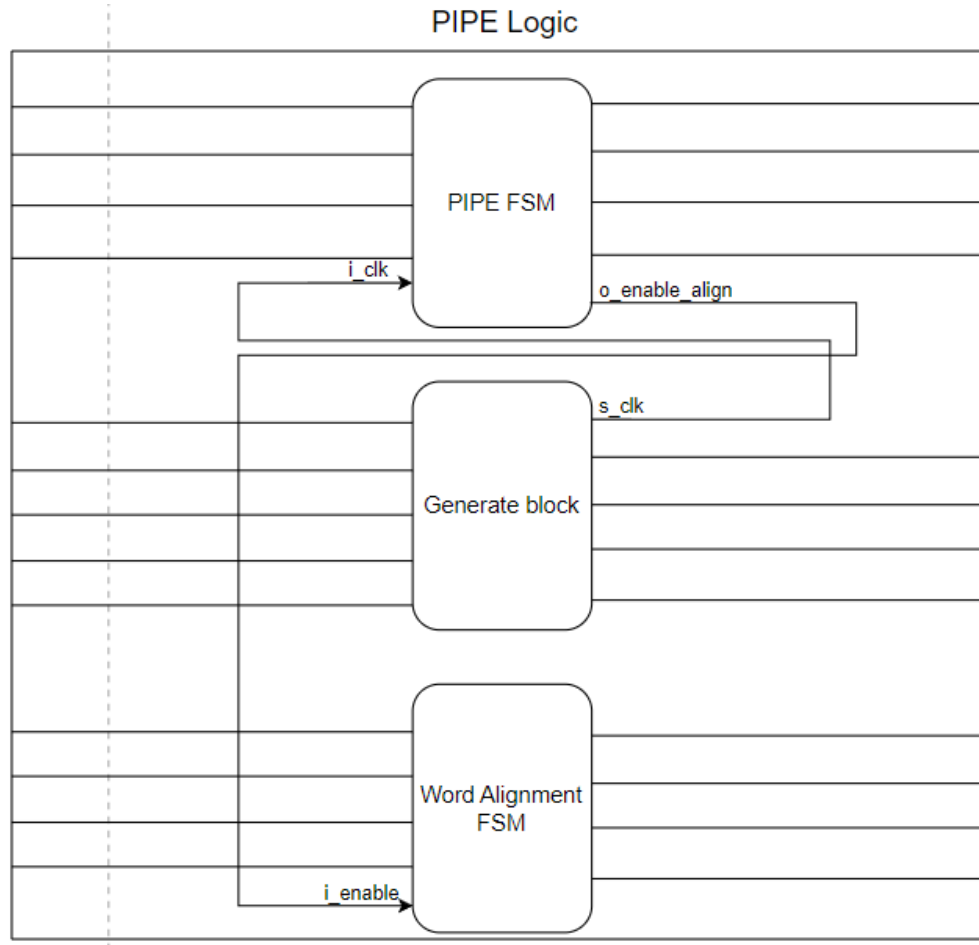
- The **PHY Interface for the PCI Express Architecture (PIPE)** is part of the physical layer and serves as an interface between the Media Access Layer (MAC) and the Physical Coding Sublayer (PCS).

PIPE Overview

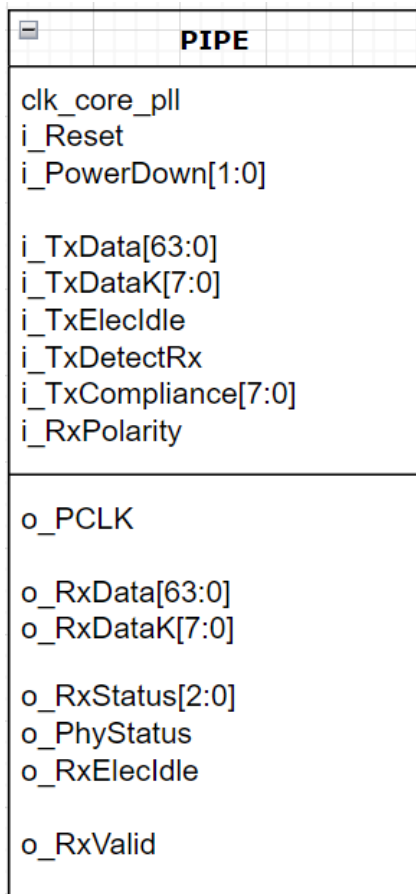


- PIPE Interface with MAC layer.
- Custom Support Signals (CSS) Interface for internal PIPE IP Core logic. These signals are mainly used for simulation purposes or to generate signals for PIPE Interface.
- Register interface for SerDes configuration.

General PIPE Logic Block diagram

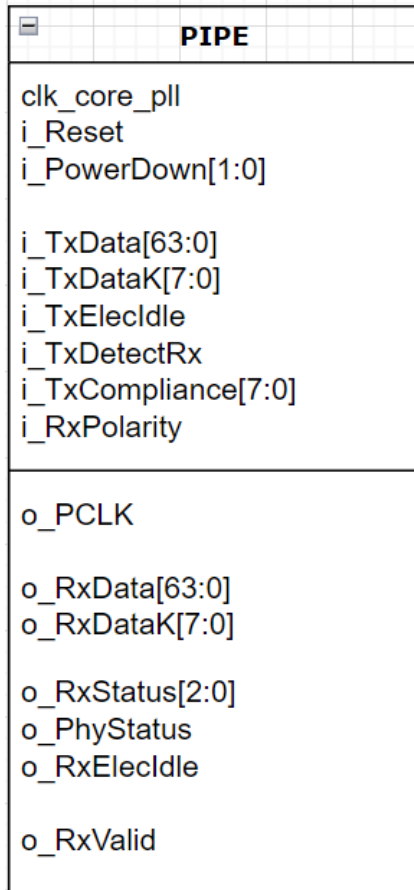


PIPE Interface



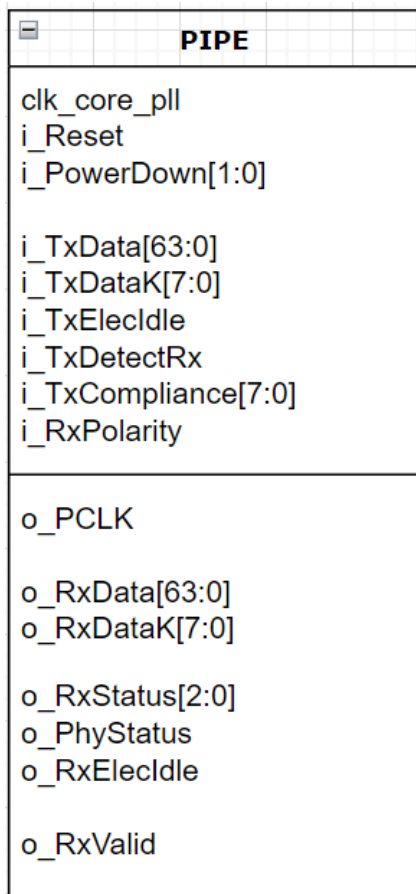
- Reference clock from ADPLL.
- Parallel Interface Clock for MAC and PHY synchronization.
- Asynchronous Reset for Transceiver.
- Power states.

PIPE Interface



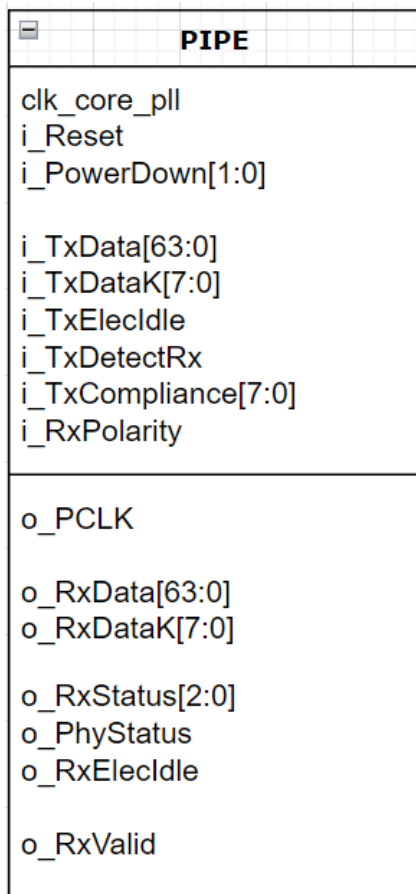
- Data from MAC for SerDes Transmitter.
- Control signals triggered by MAC layer.

PIPE Interface



- Data from SerDes Receiver.

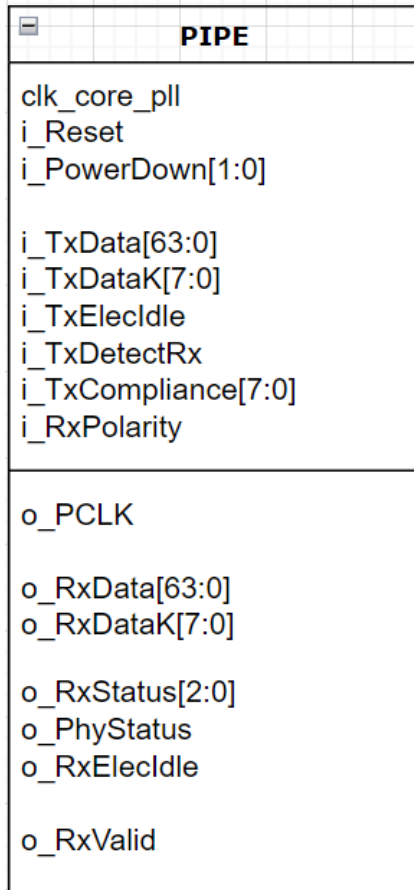
PIPE Interface



RxStatus	Description
000	Received Data OK (P0), Receiver not detected (P1)
001	SKP added
010	SKP removed
011	Receiver detected (P1)
100	8b/10b Error and optionally Disparity Error
101	Elastic Buffer Overflow
110	Elastic Buffer Underflow
111	Disparity Error. Unused if reported together with 8b/10b Error (100).

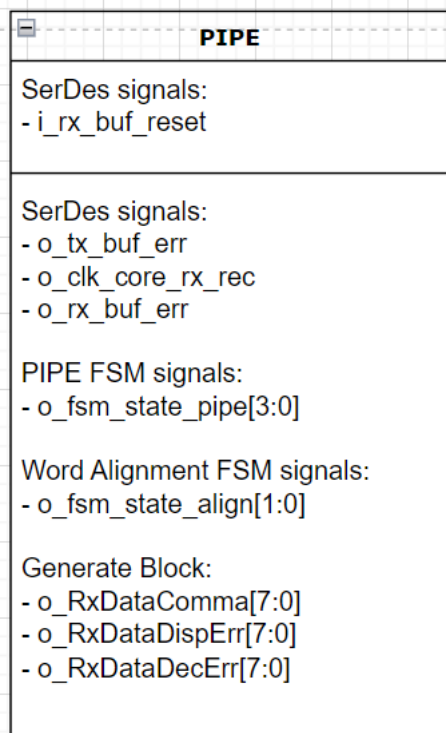
- RxStatus: Status of the received data.
- PhyStatus: Successful power state change or finished Receiver Detection.

PIPE Interface

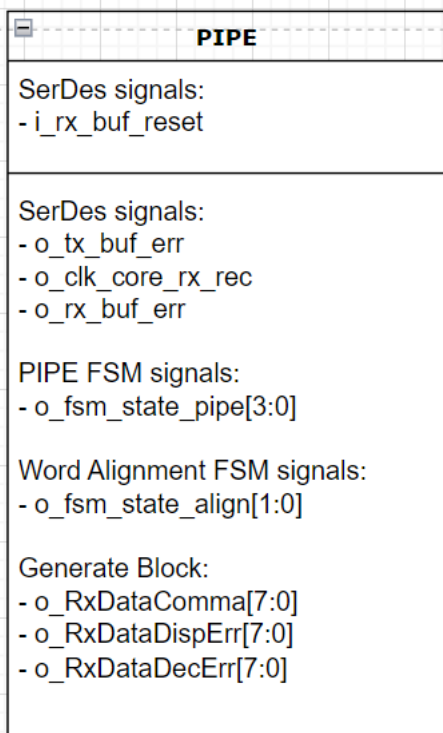


- RxValid: Indicates symbol lock/Received data is valid.

CSS Interface



- SerDes signals:
 - i_rx_buf_reset: trigger a reset for the elastic buffer.
 - o_tx_buf_err: Overflow or Underflow in Transmitter buffer.
 - o_clk_core_rx_rec: Recovered Clock from the received data.
 - o_rx_buf_err: Overflow or Underflow in Elastic Buffer of SerDes Receiver.



- **PIPE FSM:**
 - o_fsm_state_pipe [3:0]: current state of the PIPE FSM.
- **Word Alignment FSM:**
 - o_fsm_state_align [1:0]: current state of the Word Alignment FSM.
- **Generate Block:** Module that has received and transmitted data along with their error flags as inputs and generate status signals accordingly.
 - o_RxDataComma [7:0]: Each bit represents if the symbol at that position is a COM or not.
 - o_RxDataDispErr [7:0]: Each bit represents if the symbol at that position has disparity error or not.
 - o_RxDataDecErr [7:0]: Each bit represents if the symbol at that position has 8b/10b Decode error or not.

Register Interface

Register Interface
clk_reg regfile_en regfile_we regfile_addr regfile_di
regfile_do regfile_rdy

- clk_reg: reference clock for the dynamic reconfiguration port.
- regfile_en: enable control signal to access the dynamic reconfiguration feature.
- regfile_we: provides the write enable control signal to write the data on the data input port into the register selected by the address bus.
- regfile_addr: dynamic reconfiguration address.
- regfile_di: provides reconfiguration data that writes into a specified address of the register set.
- regfile_do: provides data output of the register selected by the address bus.
- regfile_rdy: provides the response to the enable signal for the PLL's dynamic reconfiguration feature. This signal is pulsed High when a write or read operation is successful.
- Register mapping can be found in [Gatemate datasheet](#) (Page 90-97).

PIPE FSM

PIPE FSM
i_pw_state[1:0] i_tx_elec_idle i_tx_detect_or_loop i_rx_detect_done i_clk
o_fsm_state_pipe[3:0] o_rx_status[2:0] o_phy_status o_rx_detect_start o_tx_elec_idle o_enable_align

- Inputs:
 - i_pw_state [1:0]: current power state.
 - i_tx_elec_idle: set transmitter in electrical idle state when set.
 - i_detect_or_loop: trigger receiver detection or loopback mode when set.
 - i_rx_detect_done: receiver detection is done.
 - i_clk: input clock.
- Outputs:
 - o_fsm_state_pipe [3:0]: current state of PIPE FSM.
 - o_rx_status [2:0]: receiver status.
 - o_phy_status: physical status flag.
 - o_rx_detect_start: initiate receiver detection.
 - o_tx_elec_idle: set transmitter in electrical idle state.
 - o_enable_align: enable word alignment.

States in PIPE FSM

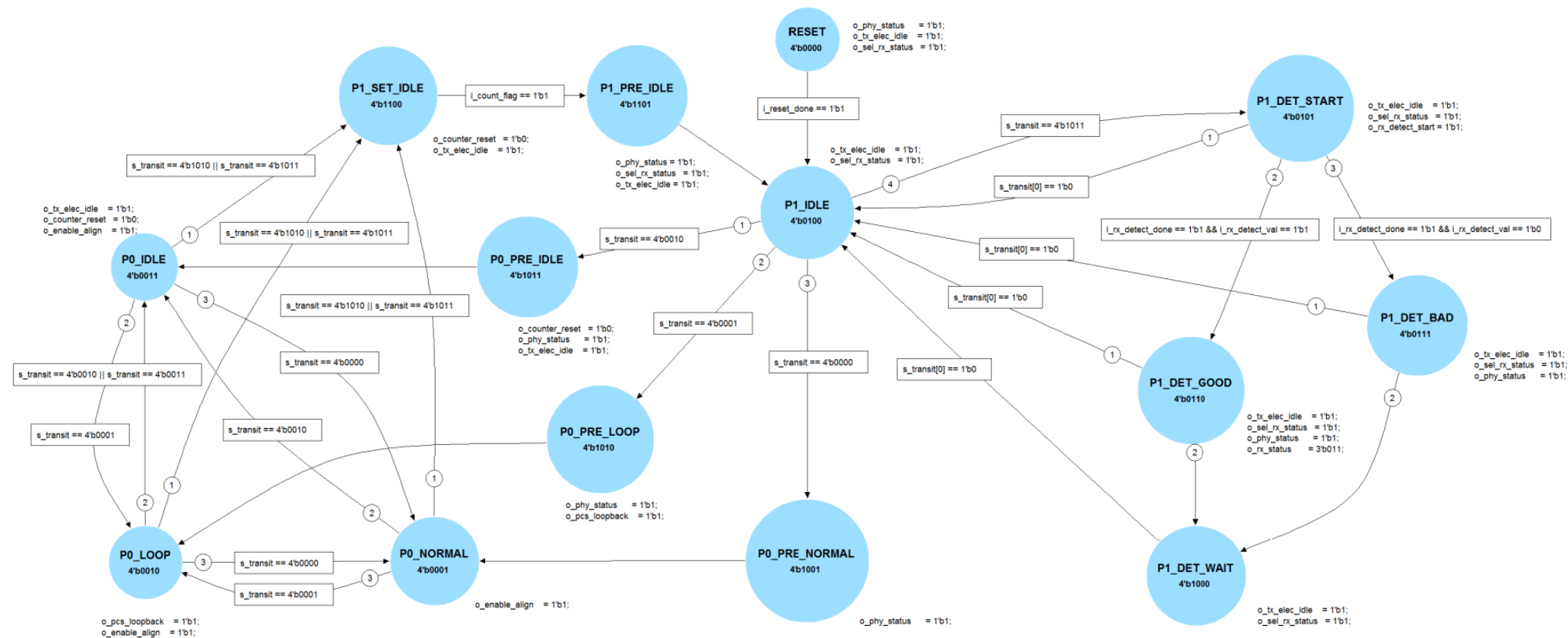
PIPE FSM	
RESET	Reset state
P1_IDLE	Transmitter in Electrical Idle state
P1_DET_START	Trigger Receiver Detection operation, set i_TxDetectRx to 1
P1_DET_GOOD	Receiver detected (RxStatus = 011)
P1_DET_BAD	Receiver not detected (RxStatus = 000)
P1_DET_WAIT	Wait for i_TxDetectRx back to 0
P0_NORMAL	Normal operation
P0_LOOP	Loopback mode
P0_IDLE	Transmitter in Electrical Idle state
P1_SET_IDLE	Trigger transition from power state P0 to P1

- Prefix Px represents the current power state.
- Each state transition where a power state change is possible would also require a PRE-state in between. This state will set the PhyStatus flag to 1'b1. The flag is deasserted when the power state change is successfully completed. The behavior follows the PIPE Specs.
- Power state change possible for:
 - From P1_IDLE to all P0 states.
 - From P0 states through P1_SET_IDLE to P1_IDLE.

States in PIPE FSM



```
s_transit = { i_pw_state , i_tx_elec_idle , i_tx_detect_or_loop }
```



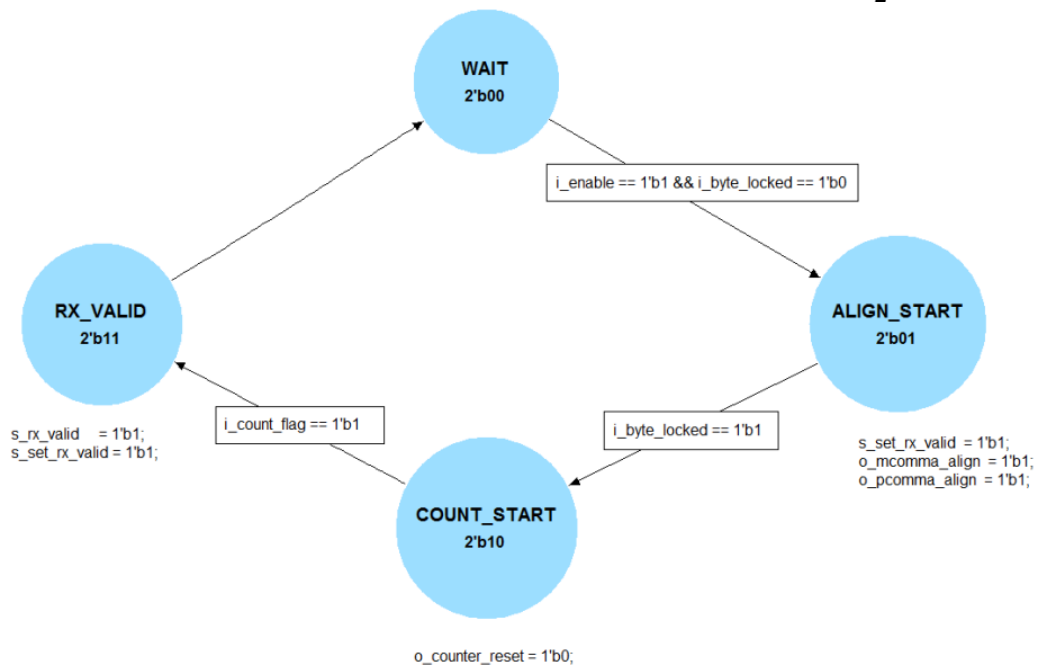
Word Alignment FSM

Word Alignment FSM
i_clk i_enable i_byte_locked
o_fsm_state_align[1:0] o_rx_valid o_mcomma_align o_pcomma_align

- Inputs:
 - i_enable: enable word alignment.
 - i_byte_locked: byte is locked.
 - i_clk: input clock.
- Outputs:
 - o_fsm_state_align [1:0]: current state of Word Alignment FSM.
 - o_rx_valid: received data is valid.
 - o_mcomma_align: activate word alignment of SerDes (negative comma sequence).
 - o_pcomma_align: activate word alignment of SerDes (positive comma sequence).

States in Word Alignment FSM

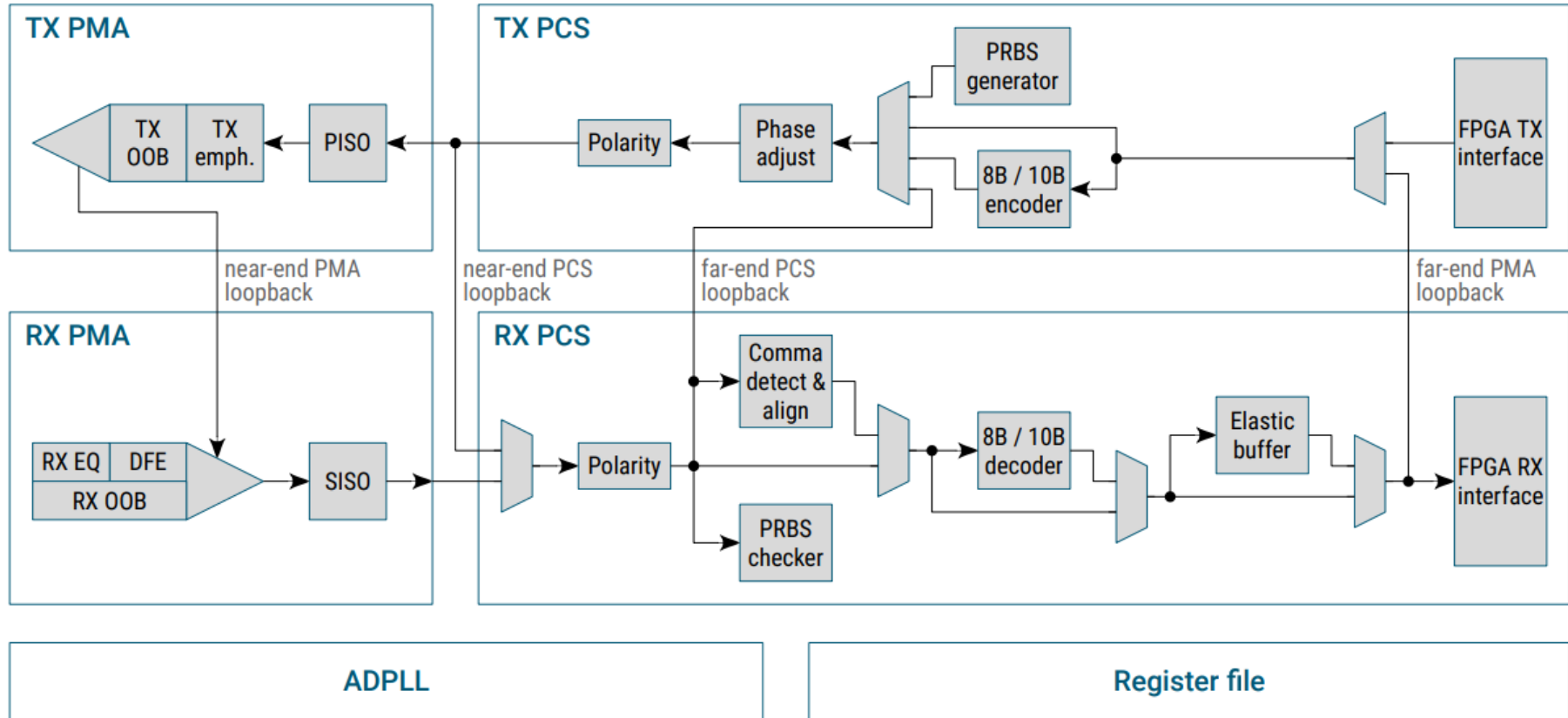
Word Alignment FSM	
WAIT	Reset state
ALIGN_START	Alignment enabled but byte is not locked
COUNT_START	Byte is locked, start counter
RX_VALID	Target value reached, received data is valid



References

- [PHY Interface for the PCI Express* Architecture \(PIPE\) Specs](#)
- [AMD DRP Definition](#)
- [Philipp Ledüç's Masterthesis](#)

Appendix A: SerDes Architecture



Appendix A: SerDes Architecture

- Physical coding sublayer (PCS)
 - Configurable 16 / 20-, 32 / 40- or 64 / 80-bit transmitter and receiver data paths.
 - 8B / 10B encoding and decoding.
 - Comma detection and byte alignment.
 - Transmitter Clock and Data Recovery (CDR).
 - Multiple pseudo-random bit stream (PRBS) generators and checkers.
 - Phase adjust FIFO for clock correction (elastic buffer).
 - Polarity control.
- Physical media attachment (PMA)
 - 3-tap decision feedback equalizer (DFE).
 - Transmitter pre- and post-emphasis.
 - Configurable transmitter driver.

Appendix A: ADPLL of SerDes

- The ADPLL of the SerDes generates the bit rate clock for the serial transmission from an external reference clock.

Name	Width	Mode	Default	Description
PLL_MAIN_DIVSEL	6	r/w	27	$[1:0] = 00 : N_2 = 3$ $[1:0] = 01 : N_2 = 2$ $[1:0] = 10 : N_2 = 4$ $[1:0] = 11 : N_2 = 5$ (default) $[2] = 0 : N_1 = 1$ (default) $[2] = 1 : N_1 = 2$ $[4:3] = 00 : N_3 = 3$ $[4:3] = 01 : \text{n.a.}$ $[4:3] = 10 : N_3 = 4$ $[4:3] = 11 : N_3 = 5$ (default) $[5] = 0 : \text{unused, do not change}$
PLL_OUT_DIVSEL	2	r/w	0	$00 : M_3 = 1$ $01 : M_3 = 2$ $10 : \text{n.a.}$ $11 : M_3 = 4$
PLL_FCNTL	6	r/w	58	Frequency divider M_1 selection for SerDes transmitter clock output f_{DPC} to FPGA core (see Table 2.17 for M_1 values)

PLL_FCNTL	M_1	PLL_FCNTL	M_1
0	*1	32	9
1	*1	33	10.5
2	3	34	9
3	3.5	35	10.5
4	3.5	36	10.5
5	3.75	37	11.25
6	4	38	12
7	4	39	12
8	4.5	40	13.5
9	5.25	41	15.75
10	5.25	42	15
11	5.5	43	16.5
12	5.5	44	16.5
13	5.75	45	17.25
14	6	46	18
15	6	47	18
16	6	48	12
17	7	49	14
18	6	50	12
19	7	51	14
20	7	52	14
21	7.5	53	15.75
22	8	54	16.5
23	8	55	16.5
24	9	56	18
25	10.5	57	21
26	10	58	20
27	11	59	22
28	11	60	22
29	11.5	61	23
30	12	62	24
31	12	63	24

BRC: Bit Rate Clock

$$f_{BRC} = f_{SER_CLK} \cdot \frac{N_1 \cdot N_2 \cdot N_3}{M_3}$$

DPC: Data Path Clock

$$f_{DPC} = f_{SER_CLK} \cdot \frac{N_1 \cdot N_2 \cdot N_3}{M_3} \cdot \frac{1}{M_1 \cdot M_4} = \frac{f_{BRC}}{M_1 \cdot M_4}$$

$M_4 = 2$ for 64-Bit SerDes data path,
otherwise $M_4 = 1$

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